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PATENTABILITY SEARCH REPORT ON

(TITLE)“METHOD AND SYSTEM FOR PROVIDING A DIGITAL JUKEBOX”

Our Ref.: KHBAJ-001

FOR

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Contents

1 Understanding of the Proposed Research Work

The present invention relates to a method and system for providing a digital jukebox. The DJ creates a session by opening JukeBoB and creates a new "jukebox session." They give it a name (e.g., "John's Wedding Reception") and sets a password to keep it secure. The platform generates a unique session code (like "ABC123") and a QR code. The DJ displays this session code so guests can join their session. The DJ views a dashboard showing: all song requests, who requested them, tip amounts, and the order songs should be played. The DJ can manage the queue by marking songs as "played" when finished, skipping songs (tips are refunded if skipped), and seeing total earnings in real-time. At any time, the DJ can "checkout" to receive their tips. JukeBoB takes a 5% platform fee, and the DJ receives the remaining 95%.

2 Key Features and Result Summary

2.1 Key Features

A method and system for providing a digital jukebox, comprising:

1. Enabling creation of a digital jukebox session for a user. The digital jukebox session creation includes giving a name and password.
2. Generating a unique session code (like "ABC123") and a QR code. A DJ displays this session code so guests can join their session.
3. Viewing a dashboard showing: all song requests, who requested them, tip amounts, and the order songs should be played.
4. Managing the queue by marking songs as "played" when finished, skipping songs (tips are refunded if skipped), and seeing total earnings in real-time.
5. At any time, the DJ can "checkout" to receive their tips. The digital jukebox collects a 5% platform fee, and the DJ receives the remaining 95%.

2.2 Result Summary

Results / Key Features	Key Feature 1	Key Feature 2	Key Feature 3	Key Feature 4	Key Feature 5
9015286	✓	✓	✓	Not explicitly disclosed	Not explicitly disclosed
20090282491	✓	✓	✓	Not explicitly disclosed	Not explicitly disclosed

3 Prior Art Search Results

3.1 Result 1

Patent Document No.	Assignee/Inventors	Date of Filing	Date of Publication
9015286	TouchTunes Music Corporation	Nov 28, 2014	April 21, 2015
Title of the Patent Document	DIGITAL DOWNLOADING JUKEBOX SYSTEM WITH USER-TAILORED MUSIC MANAGEMENT, COMMUNICATIONS, AND OTHER TOOLS		

3.1.1 Abstract from the document

A digital downloading jukebox system including a mechanism for delivering custom services to a recognized user, including services for creating playlists, communicating with others, accessing other features, etc. is provided. In some exemplary embodiments, after a user is recognized, the jukebox system allows users to access a special front-end via an Internet-enabled device or on an actual jukebox. Then, the user may, for example, create playlists, share songs with friends, send messages to friends, and access other value-added content. Such a system preferably learns about networks of friends, and enables managers to send similar messages to regular customers and/or others known to the system. In some exemplary embodiments, changes via a first user interface on a first device are reflected on second user interface on other properly-configured devices.

3.1.2 Relevant Specification

FIG. 21A shows illustrative options available for editing account/profile information in one exemplary embodiment. As noted above, it is to be appreciated that the exemplary steps illustrated in FIG. 21A may correspond to the steps required for initial account/profile setup. In such an initial setup case, however, an exemplary system might require the user to complete all of the required information at one time before allowing the user to access the main menu (step S2006 in FIG. 20). It also is to be appreciated that such account/profile information could be stored in various locations as necessary to a given implementation, such as, for example, in a central location or database, on a particular establishment's server, on specific jukebox terminals, and/or on a removable card that would identify the user and the user's information when inserted into a jukebox.

A user would first input personal data in step 2012. Such personal data may include, for example, an e-mail address for the user, a telephone (preferably mobile phone) number, and/or a name. The system also may prompt the user to create a unique username and password so that the system can identify the user at later times and from other locations. Alternatively, the system may use the user's e-mail address as a username and only require a corresponding password. Still alternatively, in the case of residential jukeboxes, for example, an exemplary system may not require a password at all. Based on the personal data, the exemplary system may gather enough information to recognize the user at logins (e.g., online access, preferred location, other locations within a given area, etc.) as well as send e-mail and text messages to the user. These exemplary features will be discussed in greater detail below.

Users may be billed according to billing information they inputted. For example, a stored credit card may be charged whenever a song is downloaded via the jukebox interface. As another example, a user may setup a special jukebox-specific account, linked to, for example, a Pay-Pal account. In a related aspect, users may use their cell phones (or other suitable portable devices) to purchase media, register with the jukebox, etc. by exchanging codes (e.g. via SMS messages) with the jukebox. For example, a user may first identify himself or herself with the jukebox (e.g. may login on the jukebox or remotely through a mobile device). Depending on whether the user's cell phone number (or e-mail address, etc., depending on the implementation), the user may have to enter the appropriate information. An SMS (or appropriately similar) authorization message and/or code may be sent to the user. The user may then enter that code on the jukebox to begin purchasing music, creating personalized content, accessing features available to recognized users, etc. This method of communication also may in some example embodiments enable users to pre-verify and/or pre-pay for music purchases remotely.

Preferably, a successful log-on displays a main menu for users. FIG. 23 is an exemplary screen shot showing the features available after a recognized user has logged in, in accordance with one exemplary embodiment. A customized greeting (e.g., the user's name) is displayed in area 2300. Playlists 2302, 2304, and 2306 are displayed with descriptive names along with the number of credits required to play a full playlist. For example, to play playlist 2302 costs 7 plays. Preferably, playlists are presumed to be public, which allows other users to view them. Playlists may be designated private, however, as is, for example, playlist 2306. Users can create additional playlists by pressing playlist creator button 2308, and they can edit already existing playlists by pressing playlist manager button 2310. The playlist manager may allow users to, for example, change existing playlists by adding and/or removing songs, deleting playlists completely, etc.

Users can manage a buddy list by adding a buddy by pressing add buddy button 2314. Alternatively, they can remove a buddy by pressing remove buddy button 2316. Users also can communicate by sending messages to their buddies by pressing send message button 2318. This feature can, in certain embodiments, for example, send SMS messages to mobile devices, send e-mails, send messages to appear when the recipient user next logs-in, etc. In still other embodiments, users can, for example, use this feature to send songs, donate credits, etc. Indicator area 2320 shows that the current user has received one message, and it identifies the sender. In other embodiments, however, other information may be displayed, a more complete inbox (e.g., with folders, forwarding, etc.) may be provided, etc. Credit indicator area 2322 provides account status. In this embodiment, it indicates that the user has previously purchased 20 plays online, and has entered enough money in a local jukebox terminal for 5 additional plays, for a grand-total of 25 potential plays.

Similarly, jukebox users may vote for particular instances of media to alter their priority in playlists. In this way, jukebox users can, for example, "battle" for control over the music to be played in a particular zone or particular zones within or among locations. It will be appreciated that this voting/battle mode may be implemented by using, for example, a dynamic queue, a priority queue, multiple queues, etc. It also will be appreciated that a jukebox could put into a voting mode automatically (e.g. at a particular time of day and/or on a particular day of the week), or it could be triggered manually. FIG. 19A is a flowchart showing an exemplary implementation of a jukebox voting process. Unlike conventional jukebox operations, or even jukebox bidding modes, a list of songs in a jukebox queue is displayed to users in step S1902. This list may be generated automatically, by operators, bar managers, patrons at a bar, etc. The list could be displayed, for example, on a jukebox, or, more preferably, on one or more stand-alone monitors. Additionally, in certain example embodiments, the list could be viewed by a plurality of mobile devices and/or terminals. Preferably, the information displayed contains at least the artists and names of coming songs, and a number associated with the priority of those songs. The list should be sorted by this number, so that the song with the most "votes" is displayed as the "next" song to be played, followed by the next highest song, etc. Step S1904 determines whether the currently playing song is over. If it is over, step S1906 removes it from the queue, and the next song is played in step S1908. In another embodiment, users could vote to stop/skip the currently playing song (or instance of media) by, for example, exceeding the number of votes the song had before the jukebox started playing.

As users see the coming songs, they will be tempted to push up the songs they like so the songs and/or push down the songs they do not like. In general, the more users who vote, the greater the ambiance of good songs. Thus, after step S1908, or in the case that the song is not over, the jukebox receives users' votes for particular songs in step S1910. Voting can be based on credits (as users buy credits), or tied to a user's account. In certain example embodiments, users must login to place a vote, and in certain example embodiments, users can vote a limited number of times. Preferably, users may vote from at least two distinct places within a location. The queue is updated based on this voting in step S1912, and the process returns to step S1902, where the displayed list is refreshed.

In certain example embodiments, the queue may be based on the total number of votes for particular songs. In certain other example embodiments, users may vote for and/or against certain songs and the “net” information may be displayed, indicating the number for and against, or merely the net result. If there are more votes against a song than for, the system can perform one or more of the following functions. For example, the jukebox simply may keep the song in the queue with a negative number of votes. Alternatively, the jukebox may keep the song in the queue with a zero or negative number of votes, but, for example, always wait until the net vote reaches at least 1 before playing the song. Still alternatively, the jukebox may drop any song that reaches zero or a negative number of votes.

3.1.3 *Relevant Figure*

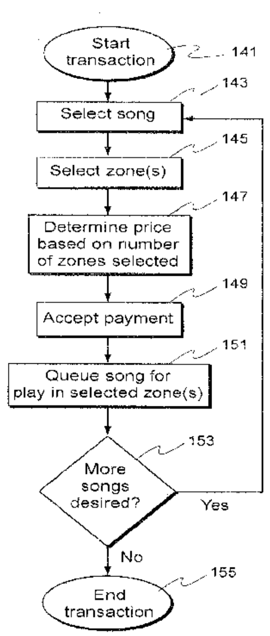

Fig. 11

3.2 Result 2

Patent Document No.	Assignee/Inventors	Date of Filing	Date of Publication
7647613	Akoo International, Inc.	July 21, 2005	Jan 12, 2010
Title of the Patent Document	Apparatus and method for interactive content requests in a networked computer jukebox		

3.2.1 Abstract from the document

A remote user interface transmits a digital media request through a communication link to a control which is coupled to a digital audio-visual playback device for selecting and playing a stored digital media by the playback device. The control receives the digital media request and transmits digital media selection and/or the selected digital media content directly to a selected playback device or indirectly to the selected playback device through a playback device server. The communication between the control and the playback device can be through a global communication network.

3.2.2 Relevant Specification

There is disclosed an apparatus and method for interactive content requests in a networked digital audio-visual playback system or computer jukebox which enables user requests for selecting a specific digital media playback at a specified playback device location to be implemented wirelessly by the user.

In one aspect, the apparatus provides for content selection in a digital audio-visual playback device capable of playing digital audio-visual media, a control means coupled in data communication with the digital audio-visual playback device for controlling at least one of a digital media transfer to the digital audio-visual playback device and a digital media selection in the digital audio-visual playback device, a communication link and a, user interface for transmitting a digital media request selecting a digital media for play by a selected digital audio-visual playback device.

In another aspect, a method for content selection in a networked digital audio-visual playback device comprises the steps of:

-
    - storing digital media in a memory;

- generating a digital media content request by a user interface remote from a digital audio-visual playback device identifying a digital audio-visual playback device and a selected stored digital media for play by the identified digital audio-visual playback device;
- transmitting the digital media content request through a communication link to a control; and
- transmitting the identified digital media by the control to the digital audio-visual playback device for play of the identified digital media by the digital audio-visual playback device.
-

The present interactive content request apparatus and method uniquely enables digital media for play through a digital audio playback device, such as a digital jukebox, to be remotely and wirelessly selected by a user. This eliminates the need for the user to move to the location of the jukebox and input a selection via the normal keyboard, mouse, and/or touchscreen display provided with prior jukeboxes. The apparatus and method also enables the user to remotely browse the available digital media content remotely from the jukebox. This can eliminate the expense of providing a display device or monitor along with the keyboard, mouse, and touchscreen inputs since the entire digital media selection process can be done remotely and wirelessly through an interface user.

The apparatus and method also provide automatic billing for each user request via any of multiple payment methods in a manner transparent to the user at the time of making a request. This eliminates the need to provide each physical digital jukebox with the monetary receiving and validation mechanism for payment of each selection.

In FIG. 1, a digital jukebox network is illustrated, by example, as including a jukebox network server 10 having a host processor which communicates via HTTP or other data transmission formats through a wireless communication system, such as the Internet 12, hardline, etc., to at least one or a plurality of digital audio-visual playback devices, such as digital jukeboxes 14. The communication format to each jukebox 14 can be via any transmission protocol, such as ADSL, cable modem, WCDMA, UMTS, or LMDS or other data communication protocols.
According to this aspect, a mobile communication device or user interface, typically a mobile wireless communication device, such as a cellular telephone 16, a PDA 18, or a computer 20, as non-limiting examples only, communicates with the communication network 22. The communication link or network 22 may be any type of communication provider either wireless or not, such as, but not limited to, WiFi, Internet, direct cable connection, Ethernet, GSM, GPRS, UMTS, EDGE, Token ring etc. Such a network 22 may have as a subset, a Short Message

Service Center (SMSC) adapted for short text messaging, an Internet protocol having displayable drop down menus allowing user input selections, or a multimedia service (MMS) providing audio and graphic data.

Although the following example of a wireless content request pertains to a short message service using an SMSC subset of the wireless carrier network 22, it will be understood that similar data input protocols or media selection, such as through drop down menus, may also be employed.

These signals are transmitted to the SMSC where the user ID is validated. The location of the digital playback device or jukebox 14, media selection, and play feature data are transmitted from the SMSC to the MMM 24 by hard wire and/or wireless communication. The MMM 24 receives the selection information from the SMSC and communicates with the jukebox network server 10 to determine the availability of the selected jukebox 14. When the server 10 has verified the corresponding jukebox 14 is operating and available to play the selected digital media, the MMM 24 sends a signal to the SMSC to bill the user for the SMS message including any premium play billing. The SMSC may then request a third party, such as, but not limited to, a bank, a credit card company, a mobile operator, etc., to debit the user account for the amount of the services procured. Other methods of payment may include m-wallet (mobile wallet), prepaid cards etc.

At the same time, the MMM 24 sends the user request along with play. criteria, such as a standard, recommendation or priority request, to the server 10. The server 10 then communicates via the Internet 12 to the selected jukebox 14 to implement the digital media selection at any premium or standard schedule.

It will be noted that the transmission of the digital media selection signals from the server 10 to the selected jukebox 14 can also include the entire selected digital media, or simply a signal to the jukebox 14 to play a particular digital media already stored in the memory of the selected venue jukebox 14.

The MMM 24, as described above, can receive a remote content or digital media selection signal from any source. The MMM 24 then checks the selection for validity, processes it, stores it in a database, and forwards the request to the network server 10 in the aspect shown in FIG. 1 or directly to the jukebox 14 in other aspects.

The server 10 or jukebox 14, 40, receives the signal from the MMM 24 and tries to the place the request to the jukebox 14, 40, etc. If this is successful, the server 10 then replies with a success message to the waiting MMM 24. If for any reason there is a failure in the request process, such as the jukebox 14 not responding, the transmission network is down, etc., the server 10 replies to the MMM 24 with a failure message along with the reason of the failure. Once the MMM 24 has received an answer to the remote content selection signal, the MMM 24 records the answer in the database and informs the user with the result of his request.

A user request from any of the mobile user interface devices 16, 18, and 20 is communicated through the network 22 to a mobile request control means 62. The user request is then communicated to the m-Venue server 61 for digital content selection at a specific digital audio-visual playback device or jukebox 14, 40, etc., or in the jukebox music server 10 which transmits the digital media selection to the individual jukeboxes 14, 40, etc., or from the content database 50 which is then transmitted by the MMM 24 directly to the jukeboxes 14, 40, etc.

Each user request is also forwarded to a reporting module 64 which is in communication with the m-Venue server 61 to record each user request and the validation of a successful digital selection. The reporting module 64 communicates with a billing module 66 which suggests the fee for each user request.

3.2.3 Relevant Figures

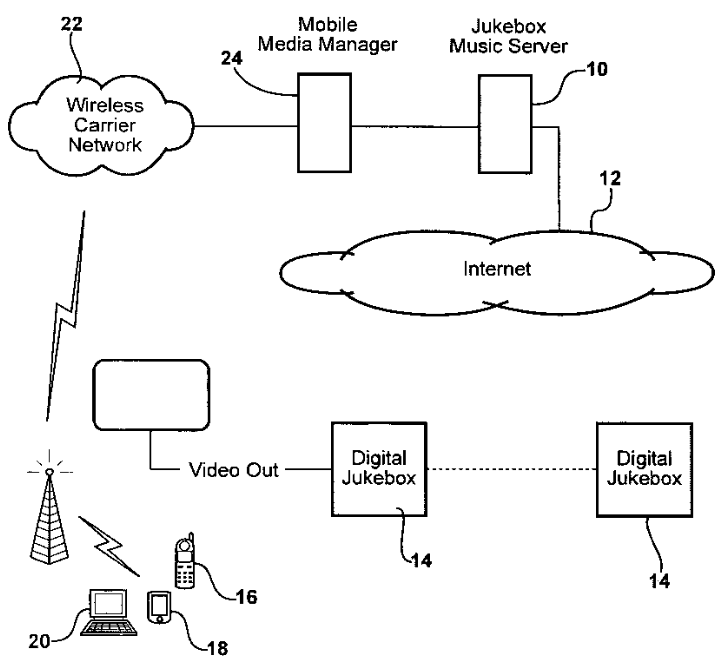

FIG - 1

3 Search Methodology

4.1 Keywords and their combinations

One or more of the keywords listed below have been used in different combinations while conducting the patentability search:

Sample Keywords/Phrases
Digital jukebox, application, session, queue, music, AI, DJ

In addition to the above mentioned key words, the following International Patent Classifications [IPCs] were also considered for conducting the search.

IPC	Description
G06F15/16	Combinations of two or more digital computers each having at least an arithmetic unit, a programme unit and a register, e.g. for a simultaneous processing of several programmes
G06F3/00	Input arrangements for transferring data to be processed into a form capable of being handled by the computer; Output arrangements for transferring data from processing unit to output unit, e.g. interface arrangements
H04N21/472	End-user interface for requesting content, additional data or services; End-user interface for interacting with content, e.g. for content reservation or setting reminders, for requesting event notification or for manipulating displayed content

4.2 Sources used

The databases/search engines used for conducting the patentability search are:

- USPTO
- Espacenet
- Patentscope [WIPO]
- Google Patents
- Freepatentsonline

<u>**CONCLUSION:**</u>

At the very outset, we would like to mention that research work is patentable, provided it meets all the three basic requirements of patentability, namely Novelty, Inventive Step and Industrial applicability, as briefly explained below:

a. ***Novelty****- the research work must have not been disclosed previously in the very same form as disclosed hereto.*

b. ***Inventive step****- the research work must be technically advanced than already existing knowledge and a person skilled in the art must not be able to arrive at the invention in view of the prior art. In other words, a research work must not be obvious to arrive at for a person skilled in the art.*

c. ***Industrial Applicability****- the research work must be of economic significance and industrially applicable.*

Documents D1 and D2 individually appear to disclose most of the critical aspects and features of the present research work. Hence, we believe that the present research work may not be <u>**novel or inventive**</u>.

Lastly, we believe that the present research work <u>satisfies</u> the requirement of ***industrial applicability***.

In view of the above, it appears that the technology disclosed in the present research work may not meet the requirement of novelty or inventive step. Hence, we believe that the present research work is <u>***not patentable***</u> by considering the features provided in the disclosure.

Let us have your comments and inputs to this search report to help us serve you better. Please feel free to contact us in case of any queries.

5 Appendix

The present opinion presents only the Attorney's view-point on the issue and does not have any legal binding attached to it. Further, the opinion is based on the search results which are found during the search. The search is carried out on documents which are made available to the public at the time of conducting the search. It should be noted that we practically do not have access to all the databases, documents, disclosures or publications globally. Further, we have taken all reasonable steps to ensure the completeness and accuracy of this report, and cannot be held liable for any error, omission or inaccuracy of the documents available in free and paid databases.

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